



**COMPARATIVE ANALYSIS OF LOGISTIC REGRESSION AND
XGBOOST FOR COGNITIVE LOAD DETECTION USING EYE
TRACKING FEATURES WITH CROSS-DATASET GENERALIZATION
EVALUATION AND FEATURE IMPORTANCE ANALYSIS**

In Partial Fulfillment of the Requirements for the
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CHAPTER 1

THE PROBLEM AND ITS BACKGROUND

PROJECT CONTEXT

Cognitive load refers to the amount of mental effort required to perform a task and is a significant factor affecting learning performance, decision-making, productivity, and human efficiency. As digital systems and technology-driven environments become increasingly complex, monitoring cognitive load has gained importance in fields such as education, healthcare, transportation, workplace systems, and human-computer interaction. Excessive cognitive load may lead to mental fatigue, decreased attention, slower reaction times, and increased human error, particularly in cognitively demanding environments [1].

In educational settings, excessive mental workload may negatively affect student comprehension, engagement, and information retention. In workplace and healthcare environments, prolonged cognitive strain may contribute to fatigue, burnout, and reduced task performance. Similarly, in safety-critical contexts such as driving, aviation, and industrial operations, cognitive overload may increase the risk of accidents and operational errors. Due to these implications, accurate and reliable cognitive load detection has become increasingly relevant for the development of adaptive, safe, and human-centered systems [2].

Traditional methods for measuring cognitive load commonly rely on subjective self-report questionnaires such as the NASA Task Load Index (NASA-TLX), as well as physiological monitoring techniques including electroencephalography (EEG), electrocardiography (ECG), and heart rate variability analysis. Although these approaches provide valuable information regarding mental workload, they often present practical limitations. Self-report methods depend heavily on participant perception and may interrupt task flow, while physiological sensors can be intrusive, costly, and difficult to implement in real-time environments [3].

Because of these limitations, researchers have increasingly explored eye-tracking technology as a non-invasive and practical alternative for cognitive load assessment. Eye-tracking enables the collection of behavioral indicators such as fixation duration, blink rate, pupil dilation, saccadic movement, and gaze patterns, which have been shown to correlate strongly with mental effort and attention [4]. With advancements in eye-tracking sensors and machine learning techniques, these features can now be analyzed automatically for cognitive state recognition.

Machine learning algorithms have demonstrated strong potential in detecting cognitive load using eye-tracking data due to their ability to identify complex patterns within behavioral datasets. Logistic Regression is commonly used as a baseline classification model because of its simplicity, interpretability, and effectiveness in binary and multiclass classification tasks. In contrast, XGBoost is a high-performance gradient boosting algorithm recognized for its robustness, predictive accuracy, and capability to model nonlinear relationships in structured data



[5]. The comparison between these two models allows evaluation between a simpler interpretable approach and a more advanced ensemble learning method for cognitive load detection.

However, despite the increasing number of studies reporting high model performance, many existing works focus primarily on single-dataset evaluation. As a result, there remains limited understanding of how well machine learning models generalize across different datasets, task conditions, and participant environments. Differences in workload scales, recording conditions, and participant behavior may affect model reliability when applied to unseen datasets. Additionally, there is still limited analysis regarding which eye-tracking features contribute most significantly to cognitive load prediction.

To address these gaps, this study conducts a comparative analysis of Logistic Regression and XGBoost for cognitive load detection using eye-tracking features. The study also evaluates cross-dataset generalization using the COLET and GAZELOAD datasets and performs feature importance analysis to identify the most influential indicators of cognitive load. By combining comparative evaluation, generalization testing, and interpretability analysis, the study aims to contribute to the development of more reliable and scalable machine learning approaches for cognitive load detection in real-world applications.

PURPOSE AND DESCRIPTION

This study aims to develop a machine learning-based framework for cognitive load detection using eye-tracking features obtained from publicly available datasets. The study specifically focuses on the comparative evaluation of Logistic Regression and XGBoost in classifying cognitive load levels using behavioral eye-tracking indicators such as fixation duration, blink rate, pupil dilation, saccadic movement, and gaze patterns. By comparing a baseline linear classification model with an advanced ensemble learning algorithm, the study seeks to determine which approach is more effective and reliable for cognitive load detection across varying task environments and participant conditions.

The proposed framework follows a structured workflow beginning with data acquisition from the COLET and GAZELOAD datasets. The collected eye-tracking data undergo preprocessing procedures including data cleaning, normalization, feature selection, and workload label harmonization to ensure compatibility between datasets. Relevant eye-tracking features are then extracted and used as inputs for the Logistic Regression and XGBoost classification models. After model training and evaluation, cross-dataset generalization testing is performed to assess how well the trained models perform on unseen datasets with different workload scales and recording conditions.

In addition to model comparison, the study also performs feature importance analysis to identify the most influential eye-tracking indicators associated with cognitive load prediction. Logistic Regression provides interpretable coefficient-based analysis, while XGBoost offers feature importance scoring through gradient-boosting mechanisms [6]. These approaches allow



the study not only to evaluate classification performance but also to examine the behavioral features that contribute most significantly to cognitive workload estimation.

The intended output of the study is a comparative machine learning framework capable of detecting cognitive load using eye-tracking features while evaluating model generalization across datasets. The findings are expected to contribute to the development of adaptive and human-centered systems in fields such as education, healthcare, workplace monitoring, and intelligent user interfaces. Furthermore, the study aims to provide insights into the applicability of interpretable and ensemble-based machine learning approaches for real-world cognitive load detection tasks [7].

OBJECTIVES OF THE STUDY

GENERAL OBJECTIVE

The general objective of this study is to compare the effectiveness of Logistic Regression and XGBoost for cognitive load detection using eye-tracking features while evaluating cross-dataset generalization and feature importance analysis.

SPECIFIC OBJECTIVES

The study aims to:

1. Preprocess and harmonize the COLET and GAZELOAD eye-tracking datasets for cognitive load analysis.
2. Extract and select relevant eye-tracking features such as fixation duration, blink rate, pupil dilation, and saccadic movement.
3. Implement Logistic Regression and XGBoost models for cognitive load classification using eye-tracking data.
4. Evaluate the performance of the selected machine learning models using accuracy, precision, recall, F1-score, and ROC-AUC.
5. Compare the classification performance of Logistic Regression and XGBoost for cognitive load detection.
6. Assess the cross-dataset generalization capability of the trained models using unseen eye-tracking data.
7. Identify the most influential eye-tracking features contributing to cognitive load prediction through feature importance analysis.



SIGNIFICANCE OF THE STUDY

This study is significant to several fields, particularly education, healthcare, workplace systems, and human-computer interaction, by contributing to the development of machine learning-based cognitive load detection using eye-tracking features. Through the comparative analysis of Logistic Regression and XGBoost, the study aims to provide insights into effective and interpretable approaches for detecting cognitive load across different environments and datasets.

In the field of education, the study may contribute to the improvement of adaptive learning systems by enabling platforms to detect when students experience cognitive overload, confusion, or disengagement. By identifying mental workload levels through eye-tracking behavior, educational systems may adjust content difficulty, pacing, and instructional strategies to support more personalized and effective learning experiences. This aligns with the United Nations Sustainable Development Goal 4 (Quality Education), which promotes inclusive and equitable quality education for all [8].

In healthcare, the study may support non-invasive monitoring of cognitive fatigue and mental strain using behavioral eye-tracking indicators such as fixation duration, blink rate, and pupil dilation. Early identification of excessive cognitive workload may contribute to mental health monitoring, cognitive assessment, and preventive interventions for individuals experiencing stress or cognitive decline. The study therefore supports Sustainable Development Goal 3 (Good Health and Well-being), which emphasizes improved mental health and well-being [9].

The study is also significant in safety-critical environments such as transportation, aviation, healthcare operations, and industrial systems. Cognitive overload is associated with increased human error, slower reaction time, and reduced decision-making performance [3]. By improving cognitive load detection, the findings may contribute to the development of systems capable of identifying excessive mental workload in high-risk tasks, thereby supporting safer and more efficient operations. This contributes to Sustainable Development Goals 3 and 11, which promote health, safety, and sustainable communities.

In the field of human-computer interaction and intelligent systems, the study may contribute to the development of adaptive interfaces capable of responding to user mental workload in real time. Eye-tracking-based cognitive load detection may improve usability, reduce user frustration, and support more efficient interaction between humans and technology. Furthermore, the comparative evaluation between Logistic Regression and XGBoost contributes to the academic and technological understanding of interpretable and ensemble-based machine learning approaches for behavioral data analysis.

The study is likewise significant to workplace and assistive systems, where cognitive load monitoring may help identify excessive workload, mental fatigue, and burnout among employees. In addition, the findings may support the development of assistive technologies for individuals with cognitive or learning difficulties by enabling systems to adapt based on user



mental state. This aligns with Sustainable Development Goals 8 and 10, which promote decent work environments and inclusive technological innovation.

Overall, this study contributes to the advancement of adaptive, safe, and human-centered systems through machine learning and eye-tracking-based cognitive load detection. By evaluating model performance, cross-dataset generalization, and feature importance, the study provides insights that may support future developments in artificial intelligence applications for education, healthcare, workplace systems, and intelligent human-centered technologies.

SCOPE AND LIMITATIONS

This study focuses on the comparative analysis of Logistic Regression and XGBoost for cognitive load detection using eye-tracking features. The study utilizes publicly available eye-tracking datasets, specifically the COLET and GAZELOAD datasets, which contain behavioral indicators associated with cognitive workload such as fixation duration, blink rate, pupil dilation, saccadic movement, and gaze patterns [11], [12].

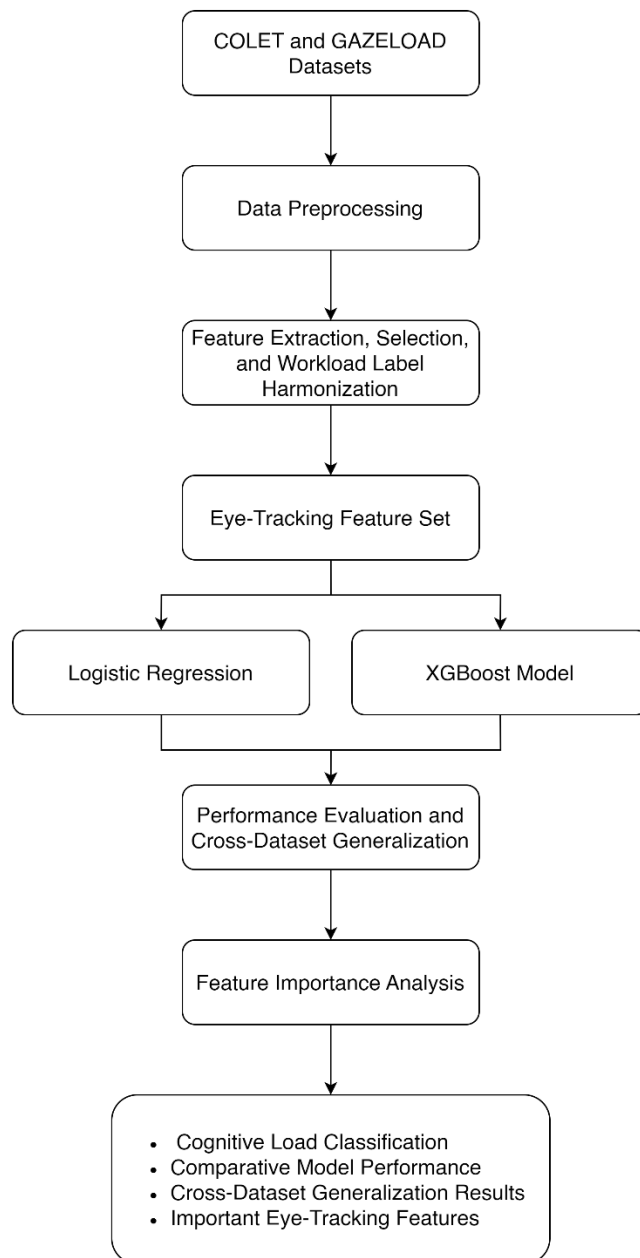
The scope of the study includes data preprocessing, feature extraction and selection, workload label harmonization, model training, performance evaluation, cross-dataset generalization testing, and feature importance analysis. Logistic Regression and XGBoost are used to classify cognitive load levels using eye-tracking features, and model performance is evaluated using standard classification metrics including accuracy, precision, recall, F1-score, and ROC-AUC [13]. The study also evaluates how well the trained models generalize across datasets with different participant groups, workload scales, and task environments.

However, the study has several limitations. The research is limited to eye-tracking data only and does not include other physiological signals such as electroencephalography (EEG), electrocardiography (ECG), heart rate variability, or facial expression analysis. In addition, the study is restricted to Logistic Regression and XGBoost and does not cover other machine learning or deep learning approaches such as neural networks or transformer-based architectures.

The study also depends on the quality, consistency, and availability of existing publicly available datasets, which may vary in terms of participant behavior, recording conditions, environmental setup, and data acquisition equipment. Since the COLET and GAZELOAD datasets use different workload labeling systems, workload labels are standardized into common cognitive load categories to support cross-dataset evaluation [14].

Furthermore, the study focuses only on offline machine learning analysis and comparative evaluation. Real-time cognitive load detection, live eye-tracking implementation, and hardware deployment are beyond the scope of this research. Despite these limitations, the study aims to provide meaningful insights into the effectiveness, generalization capability, and interpretability of machine learning approaches for cognitive load detection using eye-tracking data.

CONCEPTUAL FRAMEWORK



The conceptual framework of the study consists of two comparative machine learning models for cognitive load detection using eye-tracking features: Logistic Regression and XGBoost.

The process begins with the acquisition of publicly available eye-tracking datasets, specifically COLET and GAZELOAD, which contain behavioral indicators associated with cognitive workload. These datasets include features such as fixation duration, blink rate, pupil dilation, saccadic movement, and gaze patterns collected during cognitively demanding tasks.

The collected datasets undergo preprocessing procedures including data cleaning, normalization, workload label harmonization, and feature extraction. Since the datasets use different workload labeling systems, workload labels are standardized into common cognitive load categories to ensure compatibility during cross-dataset evaluation.

After preprocessing, the extracted eye-tracking features are used as common inputs for both the Logistic Regression and XGBoost classification models. Logistic Regression serves as a baseline interpretable machine learning approach capable of modeling linear relationships

between eye-tracking indicators and cognitive load levels. In contrast, XGBoost utilizes gradient boosting techniques to capture more complex and nonlinear feature interactions, enabling improved classification performance and feature importance analysis.

Following model training, performance evaluation is conducted using classification metrics such as accuracy, precision, recall, F1-score, and ROC-AUC. Cross-dataset generalization testing is then performed by evaluating the trained models on unseen datasets with different participant groups, task conditions, and workload scales. Feature importance analysis is also



conducted to identify the most influential eye-tracking indicators contributing to cognitive load prediction.

The framework establishes a structured comparative machine learning workflow for cognitive load detection using eye-tracking data. It provides a scalable and interpretable approach for evaluating model effectiveness and generalization capability across varying environments. Through comparative analysis and feature interpretation, the study aims to contribute to the development of adaptive, safe, and human-centered intelligent systems for real-world cognitive load monitoring applications.

OPERATIONAL DEFINITION OF TERMS

Accuracy - A performance metric used to measure the proportion of correctly classified cognitive load predictions generated by the machine learning models.

Cognitive Load - The amount of mental effort required to perform a task. In this study, cognitive load is classified using eye-tracking behavioral indicators obtained from publicly available datasets.

Cross-Dataset Generalization - The ability of a machine learning model trained on one dataset to maintain performance when evaluated on another dataset with different participant groups, workload scales, and task conditions.

Eye-Tracking - A technology used to measure eye movements and gaze-related behaviors such as fixation duration, blink rate, pupil dilation, and saccadic movement for cognitive load analysis.

Feature Extraction - The process of obtaining relevant eye-tracking indicators from the datasets to be used as input variables for machine learning classification models.

Feature Importance Analysis - A technique used to identify which eye-tracking features contribute most significantly to cognitive load prediction.

Fixation Duration - The amount of time the eyes remain focused on a specific point or object, used in this study as an indicator of mental workload.

GAZELOAD Dataset - A publicly available eye-tracking dataset containing workload-related behavioral data collected during human-robot interaction tasks, used for cross-dataset evaluation in this study.

Logistic Regression - A machine learning classification algorithm used in this study as a baseline interpretable model for cognitive load detection using eye-tracking features.

Pupil Dilation - The change in pupil size measured through eye tracking, used as an indicator of cognitive effort and mental workload.



ROC-AUC - A performance evaluation metric used to measure the ability of a classification model to distinguish between different cognitive load categories.

Saccadic Movement - Rapid eye movements between fixation points, used as one of the behavioral indicators for cognitive load analysis.

Workload Label Harmonization - The process of standardizing workload labels from different datasets into common cognitive load categories to support cross-dataset evaluation.

XGBoost - A gradient boosting machine learning algorithm used in this study for cognitive load classification and feature importance analysis using eye-tracking features.



CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

A. COGNITIVE LOAD DETECTION USING EYE-TRACKING FEATURES

As technology-driven environments become increasingly complex, understanding human mental workload has become an important area of research in education, healthcare, transportation, workplace systems, and human-computer interaction. Excessive mental workload may negatively affect attention, learning performance, decision-making, and overall task efficiency, particularly in cognitively demanding environments. Because of these implications, researchers have explored various approaches for assessing and monitoring cognitive states in real time.

One of the widely used theoretical foundations for understanding mental workload is the Cognitive Load Theory formulated by John Sweller in 1988. In it, he explains that a human's working memory can only process a limited amount of information at any given time; this amount is called cognitive load. When cognitive demands exceed its capacity, learning or task performance may deteriorate [2].

Traditional approaches for measuring cognitive load commonly rely on subjective questionnaires such as the NASA Task Load Index (NASA-TLX), as well as physiological monitoring methods including electroencephalography (EEG), electrocardiography (ECG), and heart rate variability analysis. However, these approaches may be intrusive, costly, or unsuitable for continuous real-time monitoring [3].

Recent advancements in eye-tracking technology have provided researchers with a non-invasive and behavior-based alternative for assessing cognitive load. Eye-tracking systems can capture gaze behavior and physiological indicators such as fixation duration, blink rate, pupil dilation, and saccadic movement, which have been shown to correlate with mental effort and attentional demand [1], [4]. Compared to other physiological monitoring techniques, eye tracking offers the advantage of being less physically restrictive while still providing continuous measurements of cognitive activity.

Several studies have demonstrated the effectiveness of eye-tracking features for cognitive workload recognition. Beatty (1982) found that pupil dilation increases in response to greater mental processing demands, establishing pupillary response as a reliable indicator of cognitive workload [1]. Similarly, Marshall (2007) highlighted the use of pupillometry for measuring cognitive load in human-computer interaction tasks, emphasizing its sensitivity to mental effort changes [4]. Duchowski (2017) further explained that eye-tracking metrics such as fixation patterns and gaze behavior can provide meaningful insights into user attention and cognitive processing during task performance [10].

Modern studies have expanded the application of eye tracking through machine learning approaches. Zhang et al. (2021) utilized eye movement features for cognitive workload recognition and demonstrated that gaze-related indicators significantly contribute to workload classification performance [8]. Rizzo et al. (2022) also showed that fixation, saccade, and gaze trajectory features



were effective in distinguishing between cognitive interference and non-interference conditions using machine learning classifiers. Their findings support the discriminative capability of behavioral eye-tracking indicators for cognitive state detection.

More recent studies have explored eye-tracking-based cognitive load detection in immersive and real-world environments. Nasri et al. (2024) investigated cognitive load detection during virtual reality assembly training tasks using pupil dilation and fixation duration features. Their findings demonstrated that eye-tracking indicators can effectively support machine learning-based cognitive load estimation even in complex spatiotemporal environments.

Despite the growing success of eye-tracking approaches, existing studies are often limited to controlled laboratory settings, small participant samples, or single-task experimental conditions. In addition, many studies evaluate models only within a single dataset, limiting understanding of how well cognitive load detection systems generalize across different environments and participant groups. These limitations highlight the need for further research involving cross-dataset evaluation and comparative analysis of machine learning approaches for cognitive load detection using eye-tracking features.

B. MACHINE LEARNING MODELS FOR COGNITIVE LOAD DETECTION

The selection of a machine learning model for cognitive load classification involves trade-offs among predictive performance, computational complexity, interpretability, and robustness to class imbalance and feature collinearity. Two models occupy central roles in the landscape of physiological computing research: Logistic Regression as an interpretable linear baseline and XGBoost as a high-performance nonlinear ensemble method for tabular data.

Logistic Regression models the log-odds of class membership as a linear combination of input features and produces probabilistic classification outputs amenable to direct interpretation [11]. In the context of cognitive workload detection, its coefficient vector provides an explicit mapping from gaze features to workload probability, enabling researchers to identify which features drive predictions and to test whether the learned relationships are consistent with theoretical expectations. Kaczorowska et al. [12] leveraged this interpretability to derive insights about the relative importance of fixation and pupillary features in multi-class workload discrimination, arguing that the transparency of Logistic Regression makes it methodologically superior to black-box models when the research goal includes understanding the physiological basis of workload rather than merely maximizing classification accuracy. However, Logistic Regression's assumptions of feature linearity and independence impose fundamental constraints on its expressive capacity. When gaze features exhibit nonlinear relationships with workload labels — as commonly arises from the context-dependent, multimodal nature of oculomotor responses to cognitive demands — Logistic Regression may fail to capture important discriminative structure in the data.

XGBoost, proposed by Chen and Guestrin [6], is a gradient-boosted ensemble of decision trees that learns hierarchical, nonlinear feature interactions through the sequential addition of weak learners. It has demonstrated state-of-the-art performance on tabular physiological data



across a broad range of classification tasks, owing to its capacity to model complex feature interactions, its built-in regularization mechanisms that reduce overfitting, and its robustness to missing values and feature scaling. He et al. [13] conducted a systematic comparison of Logistic Regression, XGBoost, Random Forest, SVM, and deep learning approaches using physiological signals for workload classification and found that ensemble methods, particularly XGBoost, consistently outperformed Logistic Regression in terms of balanced accuracy and F1 score across multiple evaluation conditions. Karunathilake et al. [14] similarly reported that XGBoost outperformed simpler classifiers in pupillometry-based workload prediction, attributing its advantage to the ability to exploit interactions between pupil diameter statistics and temporal fixation features that linear models cannot capture.

The 2025 study on eye-based cognitive overload prediction in human-machine interaction [15] provides the most direct antecedent for the present research, presenting a comparison of Logistic Regression and XGBoost on gaze-derived features for overload detection. XGBoost achieved the strongest predictive performance, with feature importance analysis identifying pupil-related and fixation dispersion features as the most discriminative predictors. Critically, however, that study evaluated models on a single dataset without testing whether the trained classifiers generalize to data collected under different task conditions or hardware configurations, leaving open the question of whether XGBoost's performance advantage is specific to its training distribution or reflects a generalizable inductive bias.

Rolon-Merette et al. [16] examined cross-participant workload classification using XGBoost and a Vision Transformer architecture, finding that ensemble methods offered more stable cross-subject performance than deep learning models when training data were limited, while deep learning recovered its advantage at larger dataset scales. Rizzo et al. [17] compared SVM, Random Forest, Logistic Regression, and an artificial neural network (ANN) for cognitive interference detection using fixation and saccade features, finding that ensemble and kernel-based methods offered superior performance while acknowledging that Logistic Regression retained diagnostic value through its interpretable coefficients. Collectively, these findings suggest that the performance gap between Logistic Regression and XGBoost is a function not only of model architecture but also of dataset size, feature dimensionality, and the degree to which workload-relevant signals involve nonlinear feature interactions — a relationship that may shift when models are evaluated across heterogeneous datasets rather than within a single controlled environment.

The interpretability-performance trade-off embedded in the choice between Logistic Regression and XGBoost carries methodological significance beyond classification accuracy. In applied domains such as industrial safety monitoring and clinical workload assessment, model transparency is a regulatory and ethical consideration. Post-hoc interpretability tools such as SHAP values can partially recover feature-level explanations from XGBoost models, narrowing but not eliminating the interpretability gap [18]. The 2025 overload prediction study [15] employed SHAP analysis to illuminate the feature contributions driving XGBoost predictions,



demonstrating that ensemble models can be rendered partially interpretable without sacrificing their predictive advantage.

C. CROSS-DATASET GENERALIZATION IN MACHINE LEARNING

Cross-dataset generalization is the capacity of a classifier trained on one dataset to perform accurately when applied to data collected under different conditions, it represents one of the most significant and underexplored challenges in physiological computing research. In cognitive workload detection specifically, generalization failures arise from a convergence of domain shift phenomena that include hardware variability, participant population differences, task structure heterogeneity, and ecological context divergence [19]. The proliferation of eye-tracking datasets collected under heterogeneous conditions, such as GAZELOAD and COLET, creates both an opportunity and a methodological challenge: these datasets potentially support evaluation of model robustness across real-world variation, but their differences in hardware, task design, and preprocessing introduce confounds that must be carefully controlled and analyzed.

Hardware variability represents a primary source of domain shift in eye-tracking research. Different eye trackers from fixed laboratory-grade systems such as the Tobii Pro series to wearable smart glasses such as the Meta ARIA Gen-1 used in GAZELOAD — differ in sampling frequency, gaze estimation accuracy, pupil diameter calibration, and susceptibility to lighting artifacts. These hardware-specific characteristics produce systematic differences in the statistical distributions of extracted features, even when participants perform comparable cognitive tasks [20]. Nasri et al. [20] documented how VR hardware characteristics produced feature distributions in pupil and fixation data that differed substantially from those obtained with conventional screen-based eye trackers, resulting in degraded classifier performance when models trained on one hardware configuration were applied to data from another. This hardware-induced distribution shift is particularly relevant to any comparative analysis involving GAZELOAD and COLET, which were collected using fundamentally different eye-tracking platforms in fundamentally different physical environments.

Participant variability introduces an additional layer of generalization challenge. Eye-tracking features exhibit substantial between-subject variability reflecting individual differences in baseline pupil size, blink frequency, oculomotor control, and attentional strategy [21]. Most prior ML studies in cognitive workload detection have adopted subject-dependent evaluation protocols, training and testing classifiers on data from the same participants using within-subject cross-validation [17], [12]. While this approach yields optimistic accuracy estimates, it does not assess whether the learned representations capture workload-relevant features that generalize across individuals. Rolon-Merette et al. [16] addressed this limitation by employing cross-participant evaluation protocols and demonstrated that XGBoost achieved more stable cross-subject performance than several competing approaches, though absolute performance degraded



substantially compared to within-subject conditions, highlighting the magnitude of the generalization gap.

Task structure heterogeneity compounds the difficulties of hardware and participant variability. The cognitive demands imposed by COLET's visual-search puzzle paradigm differ structurally from those of the GAZELOAD assembly task with collaborative robots. Puzzle tasks engage focused, goal-directed visual search within a constrained spatial region, producing characteristic patterns of concentrated fixations and systematic saccadic exploration [22]. Industrial assembly tasks, by contrast, require spatially distributed attention across dynamic, three-dimensional workspaces, producing fixation dispersion profiles and gaze transition entropy values that may not overlap with those generated by puzzle-solving [23]. When classifiers trained on one task type are applied to another, the mismatch between training-domain feature distributions and test-domain feature distributions constitutes a covariate shift that degrades performance even when the underlying workload states being classified are nominally equivalent. He et al. [13] noted that cross-task generalization was consistently more challenging than cross-subject generalization in their multi-condition evaluation, suggesting that task structure is a more fundamental determinant of feature distribution than individual differences.

Ecological validity further complicates the generalization landscape. Laboratory datasets such as COLET, while offering high internal validity and controlled experimental conditions, may fail to capture the cognitive complexity of naturalistic work environments. The controlled stimulus presentation and minimal physical demands of laboratory paradigms produce clean, high signal-to-noise eye-tracking data that may overstate classifier performance when models are evaluated exclusively within that context. Field datasets such as GAZELOAD, while ecologically representative, introduce noise and variability that challenge classification pipelines optimized for controlled conditions. The field of affective computing has grappled with analogous generalization challenges, and a growing literature argues that cross-dataset evaluation should be considered a standard component of model validation rather than an optional supplement to within-dataset performance reporting [19].

Despite the recognized importance of cross-dataset evaluation, the published literature on eye-tracking-based cognitive load detection contains few studies that systematically assess model performance across multiple heterogeneous datasets. Most published comparative studies, including Rizzo et al. [17] and Kaczorowska et al. [12], evaluate classifiers within a single dataset, precluding inference about generalization. The 2025 eye-based overload prediction study [15], though methodologically relevant, similarly restricts its evaluation to a single dataset, leaving the cross-dataset performance of Logistic Regression and XGBoost on gaze-derived workload features an open empirical question that directly motivates the present study.

D. RESEARCH GAPS AND SYNTHESIS



CHAPTER 3

A. RESEARCH DESIGN

This study adopts a quantitative experimental research design to compare the performance of two supervised machine learning models—Logistic Regression and XGBoost—for cognitive load detection using eye-tracking features. The research is empirical and computational, involving data preprocessing, model training, cross-dataset evaluation, and performance analysis.

A comparative approach is employed to evaluate the effectiveness of both models in accurately classifying cognitive load levels. Furthermore, cross-dataset generalization is assessed to determine how well the models perform on unseen data from a different dataset. Feature importance analysis is also conducted to identify the most significant eye-tracking indicators associated with cognitive load.

B. DATA COLLECTION METHODS

1. Eye-Tracking Dataset Collection

The study utilizes publicly available eye-tracking datasets related to cognitive load detection. These datasets consist of eye movement recordings from participants engaged in various cognitive tasks.

Each dataset includes features such as:

- Fixation duration
- Saccade amplitude
- Blink rate
- Pupil diameter
- Gaze coordinates

Each data instance is labeled according to cognitive load levels (e.g., low, medium, high), based on the experimental conditions under which the data were collected.

2. Cross-Dataset Setup

To evaluate model generalization, two separate datasets are used. One dataset is designated for training and validation, while the other is reserved for testing. This setup allows the study to assess how well the trained models generalize to unseen data from a different distribution.

C. SYSTEM ARCHITECTURE

The system consists of two main machine learning components:



Logistic Regression:

A baseline linear classification model used to predict cognitive load levels based on eye-tracking features. It is selected for its simplicity, interpretability, and efficiency in handling binary or multiclass classification tasks.

XGBoost Classifier:

An ensemble learning model based on gradient boosting, designed to improve predictive performance by combining multiple decision trees. XGBoost is chosen for its ability to handle complex, non-linear relationships and its robustness against overfitting.

D. DATA PREPROCESSING

1. Data Cleaning

Missing or inconsistent values in the datasets are handled through imputation or removal. Outliers are analyzed and treated to reduce noise and improve model performance.

2. Feature Engineering and Scaling

Numerical features are normalized or standardized to ensure consistent scale across variables.

Categorical labels (cognitive load levels) are encoded into numerical form.

Feature selection techniques may be applied to retain the most relevant eye-tracking indicators.

E. MODEL TRAINING AND EVALUATION

a. Logistic Regression

- Training Setup: Standard train-validation split applied on the training dataset.
- Metrics: Accuracy, Precision, Recall, and F1 Score.

b. XGBoost Classifier

- Training Setup: Same dataset split as Logistic Regression for fair comparison.
- Metrics: Accuracy, Precision, Recall, and F1 Score.

c. Cross-Dataset Evaluation

After training, both models are tested on a separate dataset to evaluate generalization performance. This step measures how well the models adapt to new, unseen data distributions.

F. FEATURE IMPORTANCE ANALYSIS

Feature importance analysis is conducted to identify the most influential eye-tracking features in cognitive load detection.



- For XGBoost, built-in feature importance scores are used.
- For Logistic Regression, feature coefficients are analyzed to determine the contribution of each variable.

This analysis provides insights into which eye-tracking indicators (e.g., pupil dilation, fixation duration) are most strongly associated with cognitive load.



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